

ASSIGNMENT: Module 6

* Assignment 1: Short Answer Questions

1) What is an HTML form?

→ An HTML form is a section of document used to collect input from a user, typically containing interactive controls like text input fields, checkboxes, radio buttons, or submit buttons, the collected data is then sent to a server for processing.

2) What is the purpose of the `<form>` tag?

→ The `<form>` tag is used to define an HTML form within a webpage. It acts as a container for all the form elements and specifies attributes like the action and method.

3) What is the use of the action attribute in a form?

→ The action attribute specifies the URL where the form data will be sent for processing. It defines the destination (server-side script or another webpage).

4) What is the use of the method attribute?

→ The method attribute specifies the HTTP method to be used when submitting the form data. The two most common methods (GET, POST)

5) Name any five different `<input>` types?

→ Five different input types are text, password, checkbox, radio and submit.

6) What is the role of the `<label>` tag?

→ The role of the `<label>` tag is to define a label for an `<input>` element.

7) What is the difference between `<input type = "text">` and `<textarea>`?

→ `<input type = "text">` is for single line text input.

→ `<textarea>` is for multi line text input.

8) What is the use of the `<select>` tag?

→ The `<select>` tag is used to create a drop-down list in an HTML form.

9) What is basic form validation?

→ Basic form validation is the process of ensuring that user-provided data in a web form meets necessary criteria before it is sent to server.

10) Why are forms important in real websites?

→ Forms are crucial for user interaction and data collection on websites.

Assignment 2 : Fill in the Blanks.

- 1) The tag used to create a form is <Form>
- 2) The attribute that defines where form data is sent is action
- 3) The input type used to hide text is Password
- 4) The tag used to create a dropdown list is select
- 5) The attribute used to make a field mandatory is required
- 6) The button that sends form data is called a submit button.

* Assignment 3 : Match the following element / Attribute purpose)

<Form> → Defines an HTML form for user input

<input> → Input varies based on the type attribute

<label> → defines a label for an <input> <textarea> or <select> element.

<textarea> → Multi-line text input control

<select> → drop-down list

Assignment 4: Identify the input type
Write the correct input type for the
Following fields.

1) User Name

The text input type is a single-line text field used for entering a variety of text-based data including usernames.

2) Password

The password input type masks the characters as the user types them for security purposes.

3) Email address

The Email input type is designed specifically for email addresses and provides basic client-side validation to ensure the entered text is in a valid email format.

4) Mobile number

The tel input type is for entering telephone numbers.

Assignment 8: GET vs post (conceptual)

Answer the following questions.

1) What is the GET Method?

→ The GET Method is used to retrieve data from a specified resource.

The GET method is one of the most common HTTP methods. A client uses a GET request, then asking the server to send back a specific resource, a webpage, an image or data.

2) What is the POST Method?

→ The POST Method is used to send data to a server to create or update a resource.

POST method is primary HTTP method used to submit data.

3) Which method is more secure and why?

→ Multi-Factor Authentication (MFA) MFA is more secure because it requires user to verify their identity using two or more independent factors.

4) Which method is suitable for login forms?

→ Multi-Factor Authentication (MFA) MFA is highly suitable for login forms as it provides robust security. The process typically starts with a standard username and password form, which is followed by a secondary verification step as push notification to a mobile device or entering a one time code.

5) Write one real-life use case for GET and POST each

→ * GET :- Retrieving a webpage or search results

A common real-life use case is when a user types a URL into their browser or clicks a link.

* POST :- Submitting a login form or creating a new Social media post

A typical real-life use case is submitting a login form.

Assignment 10: Real-life scenario questions

1) Where are HTML forms used on a college website?

→ college website is used student application forms for admissions, registration forms for courses or events, login page for student or faculty portals, contact us forms and feedback / survey forms.

2) Where are forms used on an e-commerce website?

→ Forms are used extensively through an e-commerce website.

Key areas include user registration and login pages, checkout process for entering shipping and billing information, search

bars for finding products, contact form for customer support, and newsletter subscription sign-ups.

3) Why is form validation important?

Form validation is important to ensure data integrity, security, and a better user experience.

4) What problems can occur if validation is not used?

Without validation, several problems can occur.

Data corruption, security risks, poor user experience, server errors, spam.